[illegible]

[illegible] :[illegible]

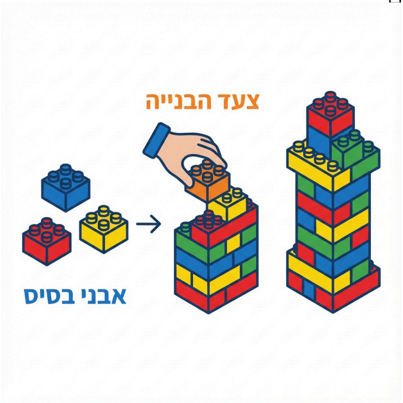
[illegible] .[illegible] :[illegible]

[illegible] / [illegible]

[illegible]: [illegible]

[illegible]: [illegible] ([illegible] → [illegible]).
[illegible] "[illegible]?", [illegible] "[illegible]?".

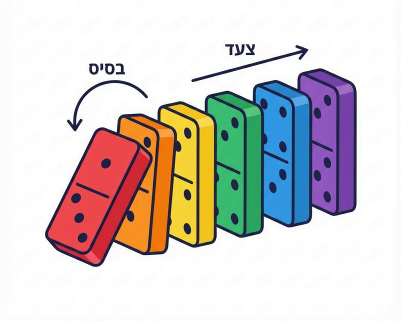
1. [illegible] (Basic Blocks): [illegible].
2. [illegible] (Construction Step): [illegible].


צעד הבנייה
אבני בסיס

[illegible]: [illegible]

[illegible]: [illegible] ($n \to n+1$ [illegible]).

1. [illegible]: [illegible].
2. [illegible]: [illegible] n [illegible] $\Longrightarrow$ [illegible] $n+1$ [illegible].
3. [illegible]: [illegible] (1, 2, 3...).



[illegible]: [illegible] [illegible] [illegible] $\mathcal{F}$

$\mathrm{Fin}(\mathbb{N})$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $\mathbb{N}$. [illegible] [illegible] $\mathcal{F}$ [illegible] [illegible] ([illegible]), [illegible] [illegible] $\mathcal{F} = \mathrm{Fin}(\mathbb{N})$:

[illegible] [illegible]

$$\emptyset \in \mathcal{F}$$

[illegible] [illegible]

[illegible] $A \in \mathcal{F}$ [illegible] $n \in \mathbb{N}$:

$$A \cup \{n\} \in \mathcal{F}$$

[illegible] [illegible] $\emptyset$

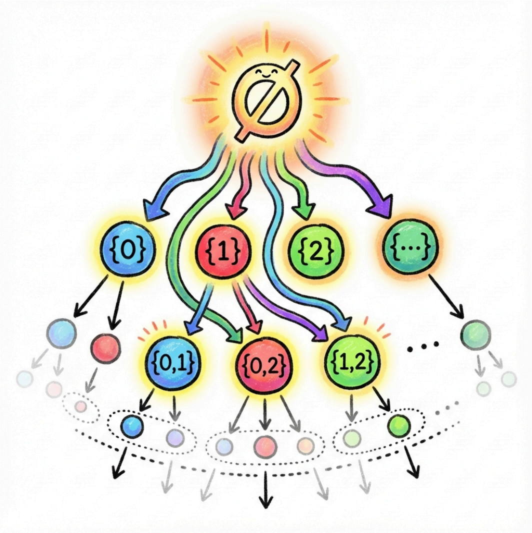
[illegible] [illegible] [illegible] - [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $\mathcal{F} \subseteq \mathrm{Fin}(\mathbb{N})$ [illegible] $\mathrm{Fin}(\mathbb{N}) \subseteq \mathcal{F}$.

$\mathcal{F}$ [illegible] [illegible] [illegible]

[illegible] [illegible] [illegible] $P(A)$ [illegible] $A \in \mathcal{F}$ [illegible]:

- $P(\emptyset)$ [illegible]
- [illegible]: $P(A) \Rightarrow P(A \cup \{n\})$ [illegible] $A \in \mathcal{F}$ [illegible] $n \in \mathbb{N}$.



[illegible] [illegible]: $\mathcal{F} \subseteq \mathrm{Fin}(\mathbb{N})$ ([illegible] [illegible] [illegible] [illegible])

[illegible]: [illegible] $A \in \mathcal{F}$, [illegible] A [illegible] [illegible] ([illegible] [illegible] [illegible] [illegible] [illegible]-$\mathbb{N}^{<k}$ [illegible] k [illegible]).

1. [illegible] ($\emptyset$):
   - [illegible] $k = 0$. [illegible] $\emptyset = \mathbb{N}^{<0}$.
   - [illegible] [illegible] ([illegible]) [illegible] [illegible] [illegible] [illegible]-$\emptyset$ [illegible]-$\mathbb{N}^{<0}$.
2. [illegible] ($A \cup \{n\}$):
   - [illegible] [illegible]-A [illegible], [illegible] [illegible] $f\colon A \to \mathbb{N}^{<k}$ [illegible] [illegible].
   - [illegible] [illegible] ($n \in A$): $A \cup \{n\} = A$, [illegible] [illegible] [illegible] ([illegible] [illegible] f [illegible]-k).
   - [illegible] [illegible] ($n \notin A$): [illegible] $g\colon A \cup \{n\} \to \mathbb{N}^{<k+1}$:

$$g(x) = \begin{cases} f(x) & x \in A \\ k & x = n \end{cases}$$

   - [illegible] [illegible] [illegible]-g [illegible] [illegible], [illegible] $A \cup \{n\}$ [illegible] ([illegible] $k+1$).

[illegible]: $\mathrm{Fin}(\mathbb{N}) \subseteq \mathcal{F}$

[illegible]: [illegible] $A \subseteq \mathbb{N}$. [illegible] A [illegible], [illegible] $A \in \mathcal{F}$.

[illegible]: [illegible] ([illegible]) [illegible] $|A|$.

[illegible]:

- [illegible] $|A| = 0$, [illegible] $A = \emptyset$.
- [illegible] $\mathcal{F}$, [illegible] $\emptyset \in \mathcal{F}$.

[illegible]:

- [illegible] k.
- [illegible] A [illegible] $k + 1$.
- [illegible] $n \in A$.
- [illegible] $A' = A \setminus \{n\}$. [illegible] $|A'| = k$.
- [illegible], $A' \in \mathcal{F}$.
- [illegible], $(A' \cup \{n\}) \in \mathcal{F}$.
- [illegible] $A' \cup \{n\} = (A \setminus \{n\}) \cup \{n\} = A$, [illegible] $A \in \mathcal{F}$.

[illegible]: [illegible], $\mathcal{F} = \mathrm{Fin}(\mathbb{N})$.

[illegible] [illegible] [illegible]

[illegible]: [illegible] S [illegible] [illegible] [illegible] B [illegible] [illegible] K.

[illegible] [illegible] $\forall x \in S, P(x)$:

1. [illegible]: [illegible] $P(b)$ [illegible] $b \in B$.
2. [illegible]: [illegible] $k \in K$, [illegible] [illegible] [illegible]:

$$P(x_1) \wedge \cdots \wedge P(x_m) \implies P(k(x_1, \ldots, x_m))$$

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible].

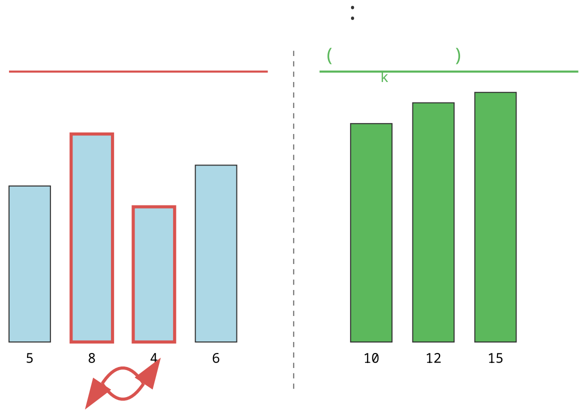
Property Preserved
Constructor
Truth
Base

[illegible]

[illegible]:

- (Lists)
- (Trees)
- [illegible] (Formulas)
- (Grammars)

[illegible].

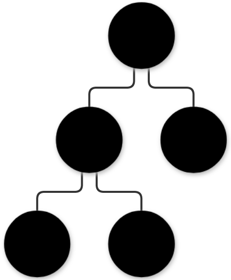

:
(
)
k
5
8
4
6
10
12
15

[illegible]

[illegible]: [illegible] $\mathcal{T}$ [illegible]:

1. [illegible]: [illegible] $\bullet$ [illegible] $\mathcal{T}$-[illegible].
2. [illegible]: [illegible] $T_1, T_2 \in \mathcal{T}$ [illegible], [illegible] $T = \text{Tree}(\bullet, T_1, T_2)$ ([illegible] $\bullet$ [illegible]-[illegible] T_1, T_2) [illegible] $\mathcal{T}$-[illegible].

[illegible]: $\text{Tree}(\bullet, \text{Tree}(\bullet, \bullet, \bullet), \bullet)$



[illegible]:

[illegible] $T \in \mathcal{T}$:

- $N(T)$ - [illegible] (Nodes).
- $L(T)$ - [illegible] (Leaves).
- $I(T)$ - [illegible] ($N(T) - L(T)$).

[illegible]:

- [illegible] ([illegible]):
- $N(\bullet) = 1, L(\bullet) = 1, I(\bullet) = 0$.
- [illegible] $T = \text{Tree}(\bullet, T_1, T_2)$:
- $N(T) = 1 + N(T_1) + N(T_2)$
- $L(T) = L(T_1) + L(T_2)$
- $I(T) = 1 + I(T_1) + I(T_2)$

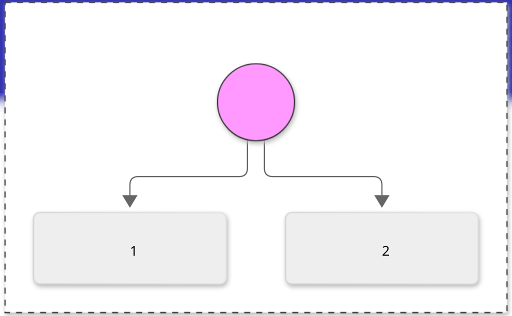
[illegible] [illegible]: [illegible] [illegible] [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] $T \in \mathcal{T}$ [illegible]: $L(T) = I(T) + 1$.

[illegible] [illegible] [illegible]:

[illegible]: [illegible] [illegible] [illegible] $T = \bullet$:

- $L(\bullet) = 1$.
- $I(\bullet) = 0$.
- [illegible]: $1 = 0 + 1$. [illegible]



[illegible]: [illegible] [illegible] [illegible] T_1, T_2. [illegible] [illegible] T [illegible] [illegible].

- $L(T) = L(T_1) + L(T_2)$ ([illegible] [illegible]).
- [illegible] [illegible]: $L(T_1) = I(T_1) + 1$ [illegible]-$L(T_2) = I(T_2) + 1$.
- [illegible]: $L(T) = (I(T_1) + 1) + (I(T_2) + 1)$.
- [illegible] [illegible]: $L(T) = (1 + I(T_1) + I(T_2)) + 1$.
- [illegible] [illegible] [illegible] [illegible]-$I(T)$:
- $L(T) = I(T) + 1$. [illegible]

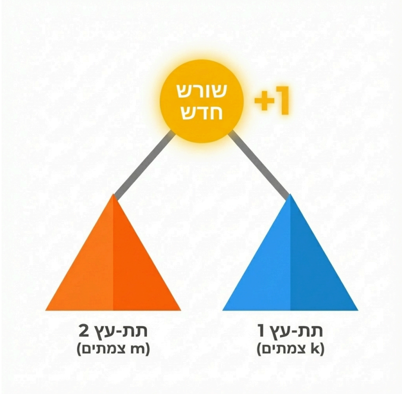
[illegible]?

[illegible]

[illegible] "[illegible]" [illegible] ([illegible]) [illegible] .([illegible]) [illegible].

- [illegible].
- [illegible] [illegible].

[illegible] -[illegible] $+1$ [illegible]!


שורש חדש
+1
תת-עץ 1
(k צמתים)
תת-עץ 2
(m צמתים)

[illegible]

- [illegible] (h):
  - [illegible] h $<$.
  - [illegible] h [illegible]-[illegible] [illegible] ([illegible] $h-1$). [illegible].
- [illegible] (n):
  - [illegible] n [illegible].
  - [illegible] "[illegible]" [illegible].

[illegible]: [illegible] [illegible] [illegible] (Propositional Logic)

[illegible]: [illegible] $Atoms$ [illegible] [illegible] [illegible] [illegible] ([illegible] $\{P, Q, R, \dots\}$). [illegible] [illegible] [illegible] [illegible] $\mathrm{PROP}(Atoms)$ [illegible] [illegible]:

1. [illegible]: [illegible] $p \in Atoms$, [illegible] $p \in \mathrm{PROP}(Atoms)$.
2. [illegible]: [illegible] $\phi \in \mathrm{PROP}(Atoms)$, [illegible] $\neg\phi \in \mathrm{PROP}(Atoms)$.
3. [illegible] [illegible]: [illegible] $\phi, \psi \in \mathrm{PROP}(Atoms)$, [illegible]:
   - $(\phi \land \psi) \in \mathrm{PROP}(Atoms)$
   - $(\phi \lor \psi) \in \mathrm{PROP}(Atoms)$
   - $(\phi \to \psi) \in \mathrm{PROP}(Atoms)$

FORMULA
BUILDING LOGIC!
P∧Q
(P∨Q)→R
P
Q
R
∧
∨
¬

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible] ([illegible] $\neg$ [illegible] [illegible]).

[illegible]: P ([illegible] [illegible]), $\neg P$, $((P \land Q) \to \neg R)$ - [illegible] [illegible].
$P \land Q$ - [illegible] [illegible] [illegible] [illegible] [illegible] ([illegible] [illegible]).

> [illegible] [illegible]: $Atoms = \{P, Q\}$, [illegible] [illegible] [illegible]: [illegible] ϕ [illegible] [illegible] $\neg\phi$ [illegible]; [illegible] ϕ, ψ [illegible] [illegible] $(\phi \land \psi)$ [illegible]. [illegible]: $(P \land Q)$, $\neg(P \land Q)$, $(\neg P \land \neg\neg Q)$.

[illegible] [illegible]: [illegible] [illegible]

[illegible]: [illegible] [illegible] $\phi \in \mathrm{PROP}(Atoms)$, [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible]: [illegible] [illegible] $T(\phi)$: $L(\phi) = R(\phi)$.

1. [illegible]: [illegible] [illegible] $p \in Atoms$:
   - $L(p) = 0, R(p) = 0$.
   - $0 = 0$ [illegible]
2. [illegible] ([illegible]): [illegible] $T(\phi)$. [illegible] [illegible] $\neg\phi$:
   - $L(\neg\phi) = L(\phi)$.
   - $R(\neg\phi) = R(\phi)$.
   - [illegible] $L(\phi) = R(\phi)$, [illegible] [illegible]. [illegible]
3. [illegible] ([illegible]): [illegible] $T(\phi)$ [illegible]-$T(\psi)$. [illegible] [illegible] $(\phi \circ \psi)$ [illegible]
   $\circ \in \{\wedge, \vee, \to\}$ ([illegible] [illegible]: [illegible] $\wedge$):
   - $L((\phi \circ \psi)) = 1 + L(\phi) + L(\psi)$.
   - $R((\phi \circ \psi)) = R(\phi) + R(\psi) + 1$.
   - [illegible] $L(\phi) = R(\phi)$ [illegible]-$L(\psi) = R(\psi)$.
   - [illegible]: [illegible] [illegible] [illegible]

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible]-$\mathrm{PROP}(Atoms)$.

([illegible]) [illegible]

[illegible]: [illegible] (True/False).

1. [illegible] (Valuation): [illegible] $v: Atoms \to \{T, F\}$ [illegible].
2. [illegible] ($\hat{v}$): [illegible] $\hat{v}: \mathrm{PROP}(Atoms) \to \{T, F\}$ [illegible]:
   - [illegible]: [illegible] $p \in Atoms$, $\hat{v}(p) = v(p)$.
   - [illegible]: $\hat{v}(\neg\phi) = T$ [illegible] $\hat{v}(\phi) = F$.
   - [illegible]:
     - $\hat{v}(\phi \wedge \psi) = T$ [illegible] $\hat{v}(\phi) = T$ [illegible] $\hat{v}(\psi) = T$.
     - $\hat{v}(\phi \vee \psi) = T$ [illegible] $\hat{v}(\phi) = T$ [illegible] $\hat{v}(\psi) = T$.
     - $\hat{v}(\phi \to \psi) = T$ [illegible] ($\hat{v}(\phi) = T$ [illegible] $\hat{v}(\psi) = F$).

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible]: [illegible] ϕ [illegible], [illegible] v_1, v_2 [illegible] [illegible]. [illegible] [illegible] [illegible] p [illegible] [illegible]-ϕ [illegible] $v_1(p) = v_2(p)$, [illegible] $\hat{v}_1(\phi) = \hat{v}_2(\phi)$.

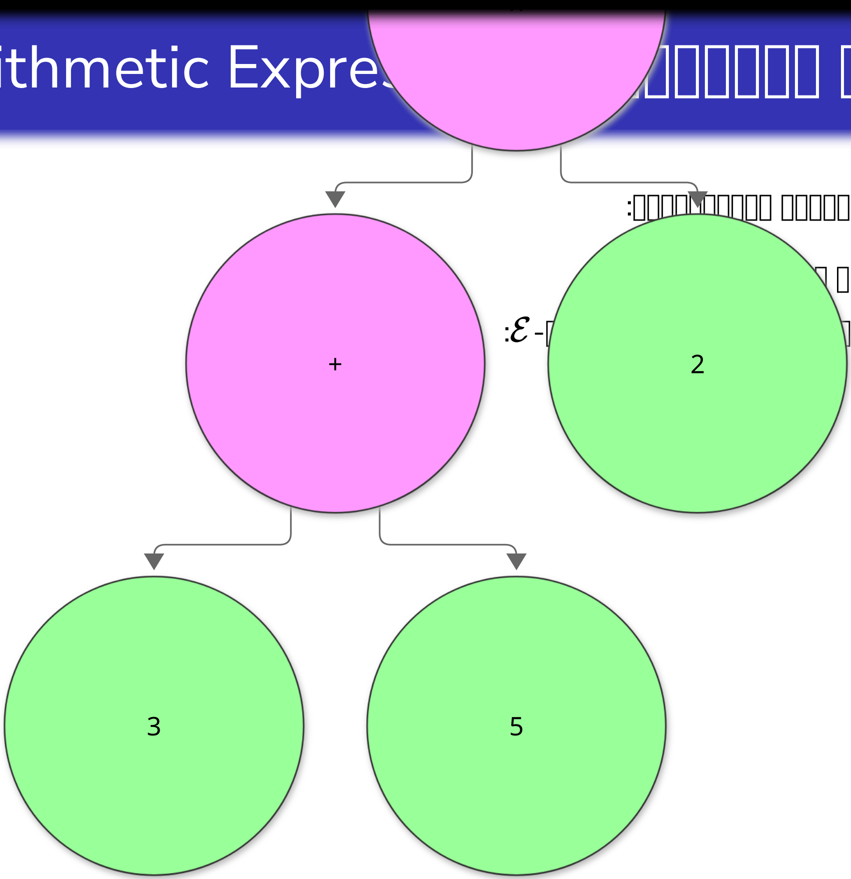
[illegible] ([illegible] [illegible] [illegible] ϕ):

1. [illegible]: $\phi = p$ ([illegible]).
   - [illegible] [illegible]-$v_1(p) = v_2(p)$.
   - [illegible], $\hat{v}_1(p) = v_1(p)$ [illegible]-$\hat{v}_2(p) = v_2(p)$. [illegible] $\hat{v}_1(\phi) = \hat{v}_2(\phi)$. [illegible]
2. [illegible] ([illegible]): $\phi = \neg\psi$.
   - [illegible]: [illegible] [illegible] [illegible] ψ. ([illegible] $\hat{v}_1(\psi) = \hat{v}_2(\psi)$ [illegible] [illegible] [illegible] ψ [illegible] [illegible] [illegible]).
   - $\hat{v}_1(\phi) = T \iff \hat{v}_1(\psi) = F \iff \hat{v}_2(\psi) = F \iff \hat{v}_2(\phi) = T$ ([illegible] [illegible]). [illegible]
3. [illegible] ([illegible]): $\phi = \psi_1 \circ \psi_2$ ([illegible] $\circ \in \{\wedge, \vee, \rightarrow\}$).
   - [illegible]: [illegible] [illegible] [illegible] ψ_1 [illegible]-ψ_2.
   - [illegible] [illegible]-ϕ [illegible] [illegible] [illegible] [illegible]-ψ_1 [illegible]-ψ_2, [illegible] [illegible] [illegible] [illegible] [illegible].
   - [illegible] [illegible] $\hat{v}(\phi)$ [illegible] [illegible] [illegible] $\hat{v}(\psi_1)$ [illegible]-$\hat{v}(\psi_2)$.
   - [illegible], [illegible] [illegible] [illegible] [illegible] v_1, v_2, [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible]

[illegible]: [illegible] [illegible] ϕ, [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

Z
X
W
R
T
T
P∧Q
רק P ו-Q חשובים

(Arithmetic Expres [illegible] :[illegible]



[illegible] $\mathcal{E}$ [illegible]:

1. [illegible]: [illegible] $n \in \mathbb{N}$ [illegible]
2. [illegible]: $E_1, E_2 \in \mathcal{E}$ [illegible] :$\mathcal{E}$-
   - $(E_1 + E_2)$
   - $(E_1 \times E_2)$

[illegible]:

- 5 ([illegible])
- $(3+5)$ ([illegible])
- $((3+5) \times 2)$ ([illegible])

[illegible]: [illegible]

[illegible] $Val\colon \mathcal{E} \to \mathbb{N}$ [illegible]:

1. [illegible]: [illegible] $n \in \mathbb{N}$, [illegible] $Val(n) = n$.
2. [illegible]:
   - $Val((E_1 + E_2)) = Val(E_1) + Val(E_2)$
   - $Val((E_1 \times E_2)) = Val(E_1) \cdot Val(E_2)$

[illegible]:

$$\begin{aligned}
Val(((3+5)\times 2)) &= Val((3+5)) \cdot Val(2) \\
&= (Val(3) + Val(5)) \cdot 2 \\
&= (3+5) \cdot 2 \\
&= 8 \cdot 2 = 16
\end{aligned}$$

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible]:

- $N(E)$ - [illegible] [illegible] ([illegible]) [illegible].
- $Op(E)$ - [illegible] [illegible] ([illegible] [illegible] [illegible]) [illegible].

[illegible]: [illegible] [illegible] $E \in \mathcal{E}$ [illegible]: $N(E) = Op(E) + 1$.

[illegible]: [illegible] [illegible] n:

- $N(n) = 1$ ([illegible] [illegible] [illegible]).
- $Op(n) = 0$ ([illegible] [illegible]).
- [illegible]: $1 = 0 + 1$. [illegible]

[illegible]: [illegible] [illegible] E_1, E_2 [illegible] [illegible] $E = (E_1 \circ E_2)$ ([illegible] $\circ \in \{+, \times\}$):

- $N(E) = N(E_1) + N(E_2)$
- $Op(E) = Op(E_1) + Op(E_2) + 1$ ([illegible] [illegible] [illegible] $\circ$).

$$\begin{aligned} N(E) &= N(E_1) + N(E_2) \\ &= (Op(E_1) + 1) + (Op(E_2) + 1) \quad \text{([illegible] [illegible] [illegible])} \\ &= (Op(E_1) + Op(E_2) + 1) + 1 \\ &= Op(E) + 1 \quad \checkmark \end{aligned}$$

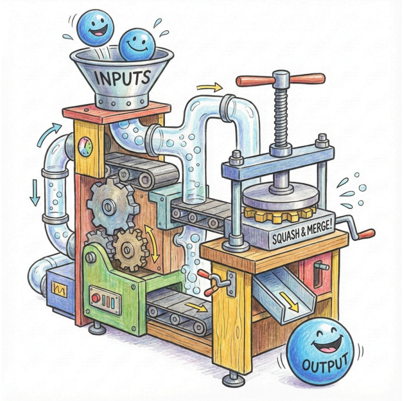
[illegible]: [illegible]

[illegible]: [illegible] E-[illegible], [illegible] $Val(E)$ [illegible].

[illegible]:

1. [illegible]: $E = n$.
   - [illegible] E-[illegible] $\Leftarrow n$ [illegible].
   - $Val(n) = n$, [illegible]. [illegible]
2. [illegible]: [illegible] E_1, E_2. [illegible] E:
   - [illegible] E-[illegible], [illegible] E_1-[illegible] E_2-[illegible].
   - [illegible]: $Val(E_1)$ [illegible] $Val(E_2)$-[illegible].
   - [illegible]: $Val(E) = Val(E_1) + Val(E_2) = \text{Even} + \text{Even} = \text{Even}$. [illegible]
   - [illegible]: $Val(E) = Val(E_1) \cdot Val(E_2) = \text{Even} \cdot \text{Even} = \text{Even}$. [illegible]

[illegible]: [illegible].


INPUTS
SQUASH & MERGE!
OUTPUT

[illegible] (FOL)

[illegible] (FOL) [illegible].

[illegible]: [illegible] "[illegible]" (p, q).

[illegible]: "[illegible]" [illegible], [illegible].

- [illegible]:

1. [illegible] (Syntax) ([illegible], [illegible], [illegible] $\forall, \exists$)
2. [illegible] ([illegible] "[illegible]" [illegible] "[illegible]"?)

[illegible] [illegible]: [illegible] (Terms)

[illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

1. [illegible] (Constants)

[illegible] [illegible] [illegible] [illegible].

- [illegible] "[illegible]" → [illegible] [illegible] [illegible].
- [illegible] "0" → [illegible] [illegible] [illegible].
- [illegible] [illegible]: a, b, c [illegible] Sarah.

2. [illegible] (Variables)

[illegible] [illegible] [illegible] [illegible] ([illegible]'[illegible]).

- "[illegible]", "[illegible]".
- [illegible] [illegible]: $v_1, v_2, \ldots$ [illegible] x, y, z.

([illegible]) [illegible]

[illegible] [illegible] [illegible]

[illegible] ([illegible]) [illegible] "[illegible]" [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

- [illegible]: [illegible] [illegible] (P, Q, R) [illegible] [illegible] ($Loves, Happy$).
- [illegible] [illegible] ([illegible]) [illegible] [illegible] (Arity) – [illegible] [illegible] [illegible] [illegible].

[illegible]:

- [illegible] 1 ([illegible]): $Happy(x)$ - "[illegible] [illegible] [illegible]" ([illegible]).
- 2 [illegible] ([illegible]): $Loves(x, y)$ - "[illegible] [illegible] [illegible] [illegible]" ([illegible]).
- 3 [illegible] ([illegible]): $Gave(x, y, z)$ - "[illegible] [illegible] [illegible] [illegible] [illegible]-[illegible]".

[illegible] :[illegible]

$R(x, y)$

[illegible], [illegible] ([illegible]).

$$Loves(\overbrace{x}^{\text{[illegible]}}, \overbrace{y}^{\text{[illegible]}})$$

[illegible]: [illegible]! $Loves(Sarah, Dan) \neq Loves(Dan, Sarah)$.

[illegible] [illegible] [illegible] [illegible] [illegible]

[illegible] [illegible] (Binary)

$$B(t_1, t_2)$$

- [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible]: $x > y$ ([illegible] [illegible]...)
- [illegible]: $Father(x, y)$ ([illegible] [illegible]...)

[illegible] [illegible] (n-ary)

$$P(t_1, t_2, ..., t_n)$$

- [illegible] ([illegible]) [illegible] [illegible] n [illegible] ($n \geq 0$).
- [illegible] $n = 1$: [illegible] [illegible] (Property).
- [illegible] $n = 0$: [illegible] [illegible] ("[illegible] [illegible]" [illegible]).

[illegible], [illegible] [illegible] [illegible]?

[illegible] [illegible] ([illegible]), [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible]!

[illegible] ([illegible]) (P)

- [illegible]: [illegible] [illegible] [illegible] (T/F).
- [illegible]: [illegible] [illegible].
- [illegible]: $Father(x, y)$
  - "x [illegible] [illegible] [illegible] y?"
  - [illegible]: [illegible]/[illegible].

[illegible] (f)

- [illegible]: [illegible] ([illegible]).
- [illegible]: [illegible] [illegible] [illegible] [illegible].
- [illegible]: $Father(y)$
  - "[illegible] [illegible] y".
  - [illegible]: "[illegible]" ([illegible] [illegible]).

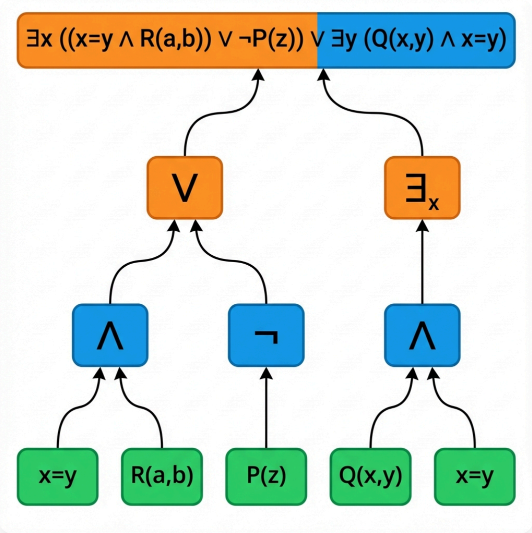
[illegible]

[illegible]. [illegible]:

1. [illegible]: $v_1, v_2, \ldots$ ([illegible] x, y, z).
2. [illegible]: R ([illegible] "[illegible]", [illegible] $\leq$).
3. [illegible]: $=$ ([illegible]).
4. [illegible]: $\neg, \vee, \wedge, \rightarrow, \exists, \forall$.

[illegible]: [illegible]

(Formulas) [illegible] [illegible]


∃x ((x=y ∧ R(a,b)) ∨ ¬P(z)) ∨ ∃y (Q(x,y) ∧ x=y)
∨
∃x
∧
¬
∧
x=y
R(a,b)
P(z)
Q(x,y)
x=y

[illegible]:

- [illegible] ([illegible]): [illegible] x, y:
  - $x = y$ [illegible].
  - $R(x,y)$ [illegible].
- [illegible]:

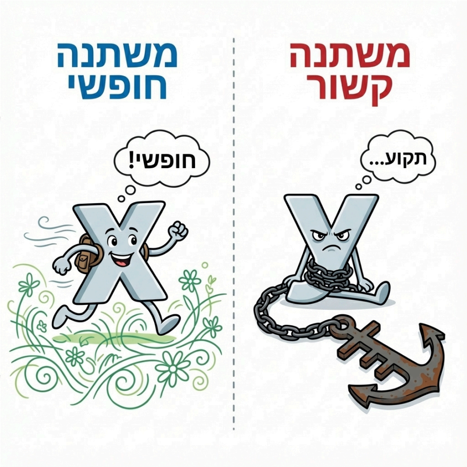
  [illegible] ϕ, ψ [illegible], [illegible]:
  - [illegible]: $\neg\phi$
  - [illegible]: $(\phi \wedge \psi), (\phi \vee \psi), (\phi \rightarrow \psi)$
  - [illegible]: $\exists x(\phi)$ [illegible]-$\forall x(\phi)$

[illegible]

[illegible] x [illegible] (Bound) [illegible] $\exists x$ [illegible] $\forall x$. [illegible], [illegible] (Free).

$$\phi_1 := R(x, y)$$

x, y . [illegible].


משתנה קשור
משתנה חופשי
תקוע...
חופשי!

$$\phi_2 := \exists y(R(x, y))$$

y ([illegible]"[illegible] ∃-[illegible]). x .

[illegible]: [illegible]-x [illegible] "[illegible]" [illegible].

$$\phi_3 := \forall x(\exists y(R(x, y)))$$

[illegible] . [illegible] "[illegible]" ([illegible]) [illegible].

[illegible] [illegible]: [illegible] [illegible]

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible], $Free(\phi)$, [illegible] [illegible]:

1. [illegible]:
   - $Free(x = y) = \{x, y\}$
   - $Free(R(x, y)) = \{x, y\}$
2. [illegible] ([illegible]): [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].
   - $Free(\neg\phi) = Free(\phi)$
   - $Free(\phi \circ \psi) = Free(\phi) \cup Free(\psi)$ ([illegible] $\circ \in \{\wedge, \vee, \to\}$)
3. [illegible] ([illegible]): [illegible] "[illegible]" [illegible] [illegible] [illegible].
   - $Free(\exists x(\phi)) = Free(\phi) \setminus \{x\}$
   - $Free(\forall x(\phi)) = Free(\phi) \setminus \{x\}$

[illegible]:

$$\begin{aligned} Free(\exists x(R(x, y) \wedge R(x, z))) &= Free(R(x, y) \wedge R(x, z)) \setminus \{x\} \\ &= (\{x, y\} \cup \{x, z\}) \setminus \{x\} \\ &= \{y, z\} \end{aligned}$$

$$\begin{aligned} Free(\forall y(x = y \to R(y, z))) &= Free(x = y \to R(y, z)) \setminus \{y\} \\ &= (\{x, y\} \cup \{y, z\}) \setminus \{y\} \\ &= \{x, z\} \end{aligned}$$

$$\begin{aligned} Free(\exists x(R(x, y)) \wedge \forall y(R(y, z))) &= Free(\exists x(R(x, y))) \cup Free(\forall y(R(y, z))) \\ &= (\{x, y\} \setminus \{x\}) \cup (\{y, z\} \setminus \{y\}) \\ &= \{y, z\} \end{aligned}$$

[illegible]: [illegible] (The Components)

[illegible] [illegible] [illegible] [illegible] ([illegible]/[illegible]), [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] M [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] [illegible] $M = \langle D, R^M \rangle$:

1. [illegible] (Domain) D: [illegible] [illegible] [illegible] [illegible] [illegible].
   - [illegible]: $D = \{1, 2, 3\}$.
2. [illegible] [illegible] [illegible] R^M: [illegible]-[illegible] [illegible] $D^2 = D \times D$ [illegible] [illegible] [illegible] [illegible].
   - [illegible]: $R^M = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$ ([illegible] "$y = x + 1$", [illegible] "x [illegible] y-[illegible] [illegible] 1-[illegible]").

[illegible] $M \models \phi[\sigma]$ [illegible]: "[illegible] ϕ [illegible] ([illegible]) M [illegible] [illegible] σ".

[illegible]:

- [illegible] [illegible]: $x \mapsto 1, y \mapsto 2 \longleftarrow M \models R(x, y)[\langle 1, 2\rangle]$ ([illegible] $\langle 1, 2\rangle \in R^M$)
- [illegible] [illegible]: $x \mapsto 2, y \mapsto 1 \longleftarrow M \models R(x, y)[\langle 2, 1\rangle]$ ([illegible] $\langle 2, 1\rangle \notin R^M$)

[illegible]: [illegible] [illegible] "[illegible]"?

[illegible] $M = \langle D, R^M \rangle$ [illegible] D ([illegible]) [illegible] $R^M \subset D \times D$.

[illegible] [illegible]. [illegible] $x_1, \ldots, x_n$ [illegible] ϕ-[illegible], $\bar{a} = \langle a_1, \ldots, a_n \rangle$-[illegible] D-[illegible]:

$$M \models \phi[\langle a_1, \ldots, a_n \rangle]$$

[illegible]: ϕ [illegible] M-[illegible] a_i [illegible] x_i.

[illegible]:

1. [illegible]: ([illegible])

- $M \models (x_1 = x_2)[\bar{a}] \iff a_1, a_2$ [illegible].
- $M \models R(x_1, x_2)[\bar{a}] \iff \langle a_1, a_2 \rangle \in R^M$.

2. [illegible]: ([illegible] $\bar{a}$)

- $M \models (\phi \wedge \psi)[\bar{a}] \Leftrightarrow$ [illegible] $\phi[\bar{a}]$ [illegible] $\psi[\bar{a}]$ [illegible].
- $M \models \neg\phi[\bar{a}] \Leftrightarrow \phi[\bar{a}]$ [illegible].
- $M \models (\phi \vee \psi)[\bar{a}] \Leftrightarrow \phi[\bar{a}]$ [illegible] $\psi[\bar{a}]$ [illegible].
- $M \models (\phi \to \psi)[\bar{a}] \Leftrightarrow$ [illegible] $\phi[\bar{a}]$ [illegible], [illegible] $\psi[\bar{a}]$ [illegible].

3. [illegible]: ([illegible] x)

- $M \models \exists x(\phi)[\bar{a}] \Leftrightarrow$ [illegible] $d \in D$ [illegible]: $M \models \phi[\bar{a}, d]$.
- $M \models \forall x(\phi)[\bar{a}] \Leftrightarrow$ [illegible] $d \in D$ [illegible]: $M \models \phi[\bar{a}, d]$.

[illegible] [illegible]

[illegible]: [illegible] M, N [illegible] ($M \cong N$) [illegible] $\pi : D_M \to D_N$ [illegible]:

$$\langle a, b\rangle \in R^M \iff \langle \pi(a), \pi(b)\rangle \in R^N$$

[illegible]: [illegible] $M \cong N$, [illegible] ϕ [illegible] σ:

$$M \models \phi[\sigma] \iff N \models \phi[\pi \circ \sigma]$$

[illegible] ([illegible]):

1. **[illegible] ($R(x, y)$):**

$$M \models R(x, y)[\bar{a}] \iff \langle a_1, a_2\rangle \in R^M \iff \langle \pi(a_1), \pi(a_2)\rangle \in R^N \iff N \models R(x, y)[\pi(\bar{a})]$$

2. **[illegible] ([illegible]):** [illegible] ([illegible], $\neg$ [illegible]).

3. **[illegible] ([illegible]):**

```
begin{aligned} M \models \exists x (\psi)[\sigma] &\iff \exists d \in D_M, M\ $$
\models \psi[\sigma[x \mapsto d]] \\ &\iff \exists d \in D_M, N \models \psi[\pi \circ
iff \exists e \in D_N, N \models\& \\ ({[illegible]}text\) \sigma[x \mapsto d]] \quad
iff N \models\& \\ ({ $pi\$ [illegible]}text\) \psi[\pi \circ \sigma[x \mapsto \pi^{-1}(e)]] \quad
$$ \exists x (\psi)[\pi \circ \sigma]. \end{aligned}
```

$$\begin{aligned} M \models \forall x(\psi)[\sigma] &\iff \forall d \in D_M, M \models \psi[\sigma[x \mapsto d]] \\ &\iff \forall d \in D_M, N \models \psi[\pi \circ \sigma[x \mapsto d]] \quad (\text{[illegible]}) \\ &\iff \forall e \in D_N, N \models \psi[\pi \circ \sigma[x \mapsto \pi^{-1}(e)]] \quad (\text{[illegible]}\ \pi\ \text{[illegible]}) \\ &\iff N \models \forall x(\psi)[\pi \circ \sigma]. \end{aligned}$$

[illegible] [illegible] :[illegible] [illegible] [illegible]

([illegible] [illegible] [illegible] [illegible] [illegible]) 26–25 [illegible] [illegible]

[illegible] [illegible] [illegible] [illegible] (L)

[illegible] L [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

1. [illegible] [illegible] [illegible] [illegible] F, [illegible] $f \in F$ [illegible] [illegible] [illegible] n_f ([illegible] (Arity) [illegible] f).
2. [illegible] [illegible] [illegible] [illegible] R, [illegible] $r \in R$ [illegible] [illegible] [illegible] n_r ([illegible] (Arity) [illegible] r).
3. [illegible] [illegible] [illegible] [illegible] C.

[illegible] [illegible]:

- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] ($=$) [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible] n-[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible]-A $f: A^n \to A$.
- [illegible] n-[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] n-[illegible] [illegible] ([illegible]-[illegible] [illegible] A^n).
- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] ($n = 2$).

[illegible] M [illegible] [illegible] L (L-Structure)

[illegible] M [illegible] [illegible] L [illegible] [illegible] [illegible] [illegible] [illegible]:

1. [illegible] [illegible] [illegible] M [illegible] , , [illegible] [illegible] [illegible] [illegible] M.
2. [illegible] $f^M : M^{n_f} \to M$ [illegible] $f \in F$.
3. [illegible] $r^M \subseteq M^{n_r}$ [illegible] $r \in R$.
4. [illegible] $c^M \in M$ [illegible] $c \in C$.

[illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] $\langle M, \dots \rangle$, [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible]:

- $L = \{R\}$ [illegible] $n_R = 2$. [illegible] $\langle \mathbb{Z}, \equiv \pmod{n} \rangle$ [illegible] L-[illegible].
- $L = \{+, \cdot, 0, 1\}$ [illegible] $+, \cdot$ [illegible] ($n_+ = n_\cdot = 2$), [illegible]-$0, 1$ [illegible].
- $\langle \mathbb{R}, +, \cdot, 0, 1 \rangle$ [illegible] L-[illegible].
- [illegible] $\langle \mathbb{R}, +, +, 0, 1 \rangle$ [illegible] L-[illegible].
- [illegible] $\langle \mathbb{R}, +, \cdot, 2, 2 \rangle$ [illegible] L-[illegible].

[illegible] [illegible] (Terms) [illegible] [illegible] L

[illegible] [illegible]-L-[illegible] (L-Terms) T [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

1. $c \in T$ [illegible] [illegible] $c \in C$.
2. $v_i \in T$ [illegible] [illegible] $v_i \in V$.
3. [illegible] $t_1, \ldots, t_{n_f} \in T$ [illegible] [illegible], [illegible] $f \in F$ [illegible] [illegible] [illegible] n_f, [illegible] $f(t_1, \ldots, t_{n_f}) \in T$.

[illegible]:

- [illegible] $L = \{+, 0\}$, [illegible] $+$ [illegible] [illegible] [illegible]-0 [illegible].
- [illegible] [illegible] [illegible] [illegible]:
  - $v_1, v_2, 0$
  - $+(v_1, v_2)$ ([illegible]: $v_1 + v_2$)
  - $((v_1 + v_2) + v_2) + 0$
- [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible]: [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] (Formulas) [illegible] [illegible] L

[illegible]: [illegible] [illegible] ϕ [illegible] [illegible] [illegible] [illegible]:

1. $t_1 = t_2$, [illegible] t_1, t_2 [illegible] [illegible] (terms-L).
2. $r(t_1, \ldots, t_{n_r})$, [illegible] $r \in R$ [illegible] [illegible] [illegible] t_i-[illegible] [illegible] [illegible].

[illegible]: [illegible] [illegible]-L-[illegible] [illegible] [illegible] [illegible] [illegible] W [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

- [illegible] $\phi \in W$, [illegible] $(\neg\phi) \in W$.
- [illegible] $\phi, \psi \in W$, [illegible] $(\phi \wedge \psi) \in W$ [illegible] $(\phi \vee \psi) \in W$.
- [illegible] $\phi \in W$, [illegible] $(\exists v_i(\phi)) \in W$ [illegible] $(\forall v_i(\phi)) \in W$.

[illegible]: [illegible] $L = \{+, 0\}$:

- [illegible] [illegible] [illegible] [illegible] [illegible]: $v_1 + v_2 = 0$.
- [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $\exists v_1(v_1 + 0 = 0)$.

[illegible] [illegible]: [illegible] [illegible] ([illegible] [illegible])

[illegible] $Var(t)$-[illegible] [illegible] [illegible] [illegible] [illegible] t. [illegible] [illegible] [illegible] [illegible] [illegible] ϕ, [illegible] $Free(\phi)$, [illegible] [illegible]:

1. [illegible] ([illegible]):
   - $Free(t_1 = t_2) = Var(t_1) \cup Var(t_2)$
   - $Free(R(t_1, \dots, t_n)) = Var(t_1) \cup \cdots \cup Var(t_n)$
2. [illegible] ([illegible]):
   - $Free(\neg \phi) = Free(\phi)$
   - $Free(\phi \circ \psi) = Free(\phi) \cup Free(\psi)$ ([illegible] $\circ \in \{\wedge, \vee, \to\}$)
3. [illegible] ([illegible]):
   - $Free(\exists x(\phi)) = Free(\phi) \setminus \{x\}$
   - $Free(\forall x(\phi)) = Free(\phi) \setminus \{x\}$

[illegible]: [illegible] ϕ [illegible] (Sentence) [illegible] $Free(\phi) = \emptyset$.

[illegible] [illegible] [illegible] [illegible] [illegible] (Term Valuation)

[illegible] L [illegible] M-[illegible] [illegible]-L. [illegible] $\sigma: V \to M$ (Assignment) [illegible].

[illegible] [illegible] [illegible] [illegible] $t^M[\sigma] \in M$ [illegible] [illegible] $t \in T$:

1. [illegible] t [illegible] [illegible] [illegible] $c \in C$, [illegible] $t^M[\sigma] = c^M$.

2. [illegible] t [illegible] [illegible] $v_i \in V$, [illegible] $t^M[\sigma] = \sigma(v_i)$.

3. [illegible] $t_1, \ldots, t_{n_f}$ [illegible] [illegible], [illegible]-$t = f(t_1, \ldots, t_{n_f})$, [illegible]:

$$t^M[\sigma] = f^M(t_1^M[\sigma], \ldots, t_{n_f}^M[\sigma])$$

[illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] (Assignment Modification)

[illegible] $\sigma\colon V\to M$ [illegible], [illegible] $v\in V$ [illegible] [illegible] $a\in M$ [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] σ, [illegible] $\sigma[v\mapsto a]$ ([illegible] $\sigma[\frac{a}{v}]$), [illegible] [illegible]:

$$\sigma[v\mapsto a](v_i)=\begin{cases} a & v_i=v\\ \sigma(v_i) & v_i\neq v\end{cases}$$

[illegible]:

- [illegible] [illegible] $L=\{+,0\}$ [illegible] [illegible] $M=\langle\mathbb{Z},+,5\rangle$ ([illegible] [illegible] 0 [illegible] [illegible] 5).
- [illegible] [illegible] σ [illegible] $\sigma(v_i)=i$.
- [illegible] $t_1=v_1+0$ [illegible]-$t_2=v_2$, [illegible]-$t=t_1+t_2$.
- [illegible] [illegible] σ:
  - $t_1^M[\sigma]=\sigma(v_1)+0^M=1+5=6$.
  - $t_2^M[\sigma]=\sigma(v_2)=2$.
  - $t^M[\sigma]=t_1^M[\sigma]+t_2^M[\sigma]=6+2=8$.
- [illegible] [illegible] $\sigma[v_2\mapsto 5]$:
  - $t_1^M[\sigma[v_2\mapsto 5]]=\sigma(v_1)+5=1+5=6$ ([illegible] [illegible]).

[illegible] [illegible] [illegible] (Satisfaction)

[illegible] M L-[illegible] σ-[illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] $M \models_\sigma \phi$ ("M [illegible] [illegible] ϕ [illegible] [illegible] σ") [illegible] [illegible] ϕ:

1. [illegible] ϕ [illegible] $t_1 = t_2$: $M \models_\sigma \phi$ [illegible] $t_1^M[\sigma] = t_2^M[\sigma]$.
2. [illegible] ϕ [illegible] $r(t_1, \dots, t_{n_r})$: $M \models_\sigma \phi$ [illegible] $(t_1^M[\sigma], \dots, t_{n_r}^M[\sigma]) \in r^M$.
3. [illegible] ϕ [illegible] $\neg\psi$: $M \models_\sigma \phi$ [illegible] [illegible] [illegible] $M \models_\sigma \psi$ ([illegible] $M \not\models_\sigma \psi$).
4. [illegible] ϕ [illegible] $(\psi \wedge \theta)$: $M \models_\sigma \phi$ [illegible] $M \models_\sigma \psi$ [illegible] $M \models_\sigma \theta$.
5. [illegible] ϕ [illegible] $(\psi \vee \theta)$: $M \models_\sigma \phi$ [illegible] $M \models_\sigma \psi$ [illegible] $M \models_\sigma \theta$.
6. [illegible] ϕ [illegible] $\exists v_j(\psi)$: $M \models_\sigma \phi$ [illegible] [illegible] $a \in M$ [illegible] [illegible]-ψ $M \models_{\sigma[v_j \mapsto a]}$.
7. [illegible] ϕ [illegible] $\forall v_j(\psi)$: $M \models_\sigma \phi$ [illegible] [illegible] $a \in M$ [illegible] $M \models_{\sigma[v_j \mapsto a]} \psi$.

[illegible] (Isomorphism)

[illegible] M, N [illegible] L.

[illegible] M-[illegible] N-[illegible] $\pi : M \to N$ [illegible]:

1. [illegible]: [illegible] $c \in C$, [illegible] $\pi(c^M) = c^N$.

2. [illegible]: [illegible] $f \in F$ [illegible] $a_1, \dots, a_{n_f} \in M$:

$$\pi(f^M(a_1, \dots, a_{n_f})) = f^N(\pi(a_1), \dots, \pi(a_{n_f}))$$

3. [illegible]: [illegible] $r \in R$ [illegible] $a_1, \dots, a_{n_r} \in M$:

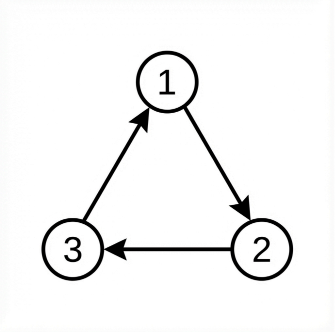
$$\langle a_1, \dots, a_{n_r} \rangle \in r^M \iff \langle \pi(a_1), \dots, \pi(a_{n_r}) \rangle \in r^N$$

[illegible], [illegible] M-[illegible] N-[illegible] $M \cong N$.

[illegible], [illegible] "[illegible]" [illegible].

[illegible]: [illegible] ([illegible])

- [illegible] L [illegible] ([illegible] R; [illegible] E).
- [illegible] L-[illegible] $M = \langle V, E\rangle$, [illegible] $E \subseteq V \times V$.
- [illegible] (Directed Graph):
  - V [illegible].
  - E [illegible].
- [illegible]:
  - [illegible] (Reflexivity): $\forall x(E(x,x))$
  - [illegible] (Symmetry): $\forall x(\forall y(E(x,y) \rightarrow E(y,x)))$
  - [illegible]-[illegible] (Anti-Symmetry): $\forall x(\forall y((E(x,y) \wedge E(y,x)) \rightarrow x = y))$
  - [illegible] (Transitivity): $\forall x(\forall y(\forall z((E(x,y) \wedge E(y,z)) \rightarrow E(x,z))))$
  - [illegible] (Linearity): $\forall x(\forall y(E(x,y) \vee E(y,x)))$



$M = \langle\{1,2,3\},\{\langle 1,2\rangle,\langle 2,3\rangle,\langle 3,1\rangle\}\rangle$

[illegible]

[illegible], $G = \langle V, E\rangle$ [illegible]-$H = \langle V', E'\rangle$,
[illegible].

[illegible]:

1. [illegible] $f : V \to V'$.
2. [illegible] $u, v \in V$:
    - $\langle u, v\rangle \in E$ [illegible] $\langle f(u), f(v)\rangle \in E'$.

[illegible]:

- [illegible].
- [illegible].
- [illegible]

[illegible]

[illegible] $G = \langle V, E\rangle$ [illegible] $H = \langle V', E'\rangle$ [illegible], [illegible]: G [illegible] H [illegible].

[illegible]:

- [illegible] H [illegible] G [illegible]:
  - [illegible] f [illegible] G-[illegible] H.
  - [illegible] $\langle u, v\rangle \in E$.
  - [illegible], $\langle f(u), f(v)\rangle \in E'$.
  - [illegible] H-[illegible], $\langle f(v), f(u)\rangle \in E'$.
  - [illegible], $\langle v, u\rangle \in E$.
- [illegible] G [illegible] H [illegible]:
  - [illegible] $g = f^{-1}$ [illegible] H-[illegible] G.

[illegible]-[illegible]

[illegible] $G = \langle V, E\rangle$ [illegible]-$H = \langle V', E'\rangle$ [illegible], [illegible]: G [illegible]-[illegible] [illegible] H [illegible]-[illegible].

[illegible]:

- [illegible] H [illegible]-[illegible] G [illegible]-[illegible]:
  - [illegible] f [illegible] G-[illegible] H-[illegible].
  - [illegible] $u, v \in V$ [illegible] $\langle u, v\rangle \in E$-[illegible] $\langle v, u\rangle \in E$.
  - [illegible], $\langle f(u), f(v)\rangle \in E'$ [illegible] $\langle f(v), f(u)\rangle \in E'$.
  - [illegible] H-[illegible]-[illegible], $f(u) = f(v)$.
  - [illegible] f-[illegible]"[illegible], $u = v$.
- [illegible] G [illegible]-[illegible] H [illegible]-[illegible]:
  - [illegible] $g = f^{-1}$ [illegible] H-[illegible] G-[illegible].

[illegible]: [illegible] (Double Induction)

[illegible]: [illegible] $m+n=n+m$ ([illegible]).

[illegible]:

1. $x+0=x$
2. $x+S(y)=S(x+y)$ [illegible] S [illegible].

[illegible] n [illegible] n [illegible] m [illegible] $m+n=n+m$.

1. [illegible] $(n=0)$: [illegible] $m+0=0+m$.
   - [illegible]: $m+0=m$ ([illegible]).
   - [illegible]: $0+m=\ldots$ [illegible]!
   - [illegible] $0+m$-[illegible].
   - ⇐ [illegible] m [illegible] $0+m=m$.

[illegible] ([illegible])

1. [illegible] 1: $0+m=m$.
   - [illegible] $(m=0)$: $0+0=0$.
   - [illegible] $(S(k))$: $0+S(k)=S(0+k)=S(k)$. [illegible]
2. [illegible] 2: $S(n)+m=S(n+m)$.
   - [illegible] $(m=0)$: $S(n)+0=S(n)=S(n+0)$.
   - [illegible] $(S(m))$:

$$S(n)+S(m)=S(S(n)+m)\overset{\text{IH}}{=}S(S(n+m))=S(n+S(m)).$$

[illegible] ([illegible]): [illegible] $m+S(n)=S(n)+m$.

$$m+S(n)\overset{\text{def}}{=}S(m+n)\overset{\text{IH}}{=}S(n+m)\overset{\text{Lemma 2}}{=}S(n)+m$$